

Diffusion for Generalized Gaussian Scale Mixtures. Geometric Target Trajectories with Learned Generative Diffusion and Flow Matching Dynamics. Applications to Image Restoration

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27-7-2026

Abstract

We provide a mathematical framework that combines Generalized Gaussian Scale Mixture (GGSM) models with modern generative techniques like Diffusion and Flow Matching.

We introduces a method to represent natural image statistics through latent precision matrices and linear transform operators, allowing for analytical solutions that do not require training a neural network from scratch for every task.

By integrating these statistical priors with pre-trained generative models, we create geometric density mixtures that balance learned semantic data with structured sparsity constraints. Several algorithms are proposed, including stochastic and deterministic sampling, to execute this integration.

Finally, the framework is applied to deblurring. A Plug-and-Play (PnP) mechanism is proposed to enforce physical consistency with observed data through a fidelity penalty during the image reconstruction process.

1 Preliminaries on Image Restoration

Let $\mathbf{x} \in \mathbb{R}^D$ denote a clean image, $\mathbf{y} \in \mathbb{R}^D$ the observed blurry image, $\mathbf{h} \in \mathbb{R}^{K \times K}$ the blur kernel (possibly unknown), and $\mathbf{n} \in \mathbb{R}^D$ the additive Gaussian noise, assumed to follow the distribution $\mathcal{N}(\mathbf{0}, \sigma^2 \mathbf{I})$.

Convolution is performed in the image space, i.e., $\mathbf{h} * \mathbf{x} = \mathbf{H}\mathbf{x}$, where $\mathbf{H} \in \mathbb{R}^{D \times D}$ is the convolution matrix corresponding to the blur kernel. The corresponding likelihood is

$$p(\mathbf{y} \mid \mathbf{x}, \mathbf{h}) \propto \exp\left(-\frac{1}{2\sigma^2} \|\mathbf{y} - \mathbf{h} * \mathbf{x}\|^2\right), \quad (1)$$

where $\|\cdot\|$ denotes the euclidean norm.

2 Preliminaries on Gaussian Scale Mixture Models

When a high-pass filter is applied to a natural image, the resulting coefficients are sparse. Super-Gaussian (SG) priors have been extensively used to model this image behavior. They have the form

$$f(x) \propto \exp(-\rho(x)) \quad (2)$$

where the function $\rho(\sqrt{s})$ has to be increasing and concave for $s \in (0, \infty)$. This condition is equivalent to $\rho'(s)/s$ being decreasing on $(0, \infty)$.

Here, we are interested in SG densities that allows for a Gaussian Scale Mixture (GSM) representation.

Formally, a density $f: \mathbb{R} \rightarrow \mathbb{R}$ admits a GSM representation if there exists a density $\pi(\omega): \mathbb{R} \rightarrow \mathbb{R}$ such that,

$$f(x) = \int_0^{+\infty} \mathcal{N}(x | 0, \omega^{-1}) \pi(\omega) d\omega. \quad (3)$$

Here, π is called the *mixing* density.

Next, we define a prior for the latent clean image starting from a family of GSM distributions. We consider a set of N high-pass filters in matrix form $\{\mathbf{F}^{(1)}, \dots, \mathbf{F}^{(N)}\} \subset \mathbb{R}^{D \times D}$. We write $\mathbf{x}^{(n)} = \mathbf{F}^{(n)} \mathbf{x}$ for $n \in \{1, \dots, N\}$. Denoting the components of $\mathbf{x}^{(n)}$ by $x_i^{(n)}$, we define,

$$p(\mathbf{x}) \propto \prod_{n=1}^N \prod_{i=1}^D f(x_i^{(n)}), \quad (4)$$

where f is a density that admits a GSM representation. We consider four different GSM densities, obtained using their SG representations $f(x) \propto \exp(-\rho(x))$, see Table 1

Name	$\rho(x)$
sech	$-\log \text{sech}(x)$
log	$\log(\epsilon + x)$
ℓ_1	$ x $
ℓ_2	$0.5 x ^2$

Table 1: GSM densities considered in this work. All are obtained as $f(x) \propto \exp(-\rho(x))$.

In summary, we have the for $\mathbf{x}$

$$p(\boldsymbol{\Omega}) = \prod_{n=1}^N \prod_{i=1}^D \pi(\omega_i^{(n)}), \quad (5)$$

$$p(\mathbf{x} | \boldsymbol{\Omega}) \propto \prod_{n=1}^N \prod_{i=1}^D \mathcal{N}(x_i^{(n)} | 0, (\omega_i^{(n)})^{-1}) \propto \mathcal{N}(\mathbf{x} | \mathbf{0}, \mathbf{P}^{-1}(\boldsymbol{\Omega})) \quad (6)$$

$$\propto \exp \left[-\frac{1}{2} \sum_{n=1}^N \mathbf{x}^\top (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \mathbf{x} \right] \quad (7)$$

where $\boldsymbol{\Omega}^{(n)} = \text{Diag}(\omega_1^{(n)}, \dots, \omega_D^{(n)})$, $\boldsymbol{\Omega} = \{\boldsymbol{\Omega}^{(1)}, \dots, \boldsymbol{\Omega}^{(N)}\}$, $\mathbf{P}(\boldsymbol{\Omega}) = \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)}$

This model corresponds to a Generalized Gaussian Scale Mixture (GGSM) with multiple linear transform operators $\mathbf{F}^{(n)}$ ($n = 1, \dots, N$).

3 Scores Identities and Langevin Simulation for GGSM Models

Let's start by calculating the marginal score. Differentiating the marginal distribution we have

$$\nabla_{\mathbf{x}} p(\mathbf{x}) = \int \nabla_{\mathbf{x}} p(\mathbf{x} | \boldsymbol{\Omega}) p(\boldsymbol{\Omega}) d\boldsymbol{\Omega} = - \sum_{n=1}^N \int (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \mathbf{x} p(\mathbf{x} | \boldsymbol{\Omega}) p(\boldsymbol{\Omega}) d\boldsymbol{\Omega} \quad (8)$$

Dividing both sides by $p(\mathbf{x})$ and using Bayes' rule $p(\boldsymbol{\Omega} | \mathbf{x}) = \frac{p(\mathbf{x} | \boldsymbol{\Omega}) p(\boldsymbol{\Omega})}{p(\mathbf{x})}$, we obtain

$$\begin{aligned}
\nabla_{\mathbf{x}} \log p(\mathbf{x}) &= \frac{\nabla_{\mathbf{x}} p(\mathbf{x})}{p(\mathbf{x})} = - \sum_{n=1}^N \int (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \mathbf{x} \frac{p(\mathbf{x}|\boldsymbol{\Omega})p(\boldsymbol{\Omega})}{p(\mathbf{x})} d\boldsymbol{\Omega} \\
&= - \sum_{i=1}^N (\mathbf{F}^{(i)})^\top \mathbb{E}[\boldsymbol{\Omega}^{(i)}|\mathbf{x}] \mathbf{F}^{(i)} \mathbf{x}
\end{aligned} \tag{9}$$

Let's now calculate $\mathbb{E}[\boldsymbol{\Omega}^{(n)}|\mathbf{x}]$. First we recall that $x_i^{(n)}$ is a GSM, which as we have indicated is also a SG model. So we have,

$$\log p(x_i^{(n)}) = -\rho(x_i^{(n)}) + \text{const} \tag{10}$$

Furthermore, for being a GSM we can write

$$\mathbb{E}[\omega_i^{(n)} | x_i^{(n)}] = - \frac{\nabla_{x_i^{(n)}} \log p(x_i^{(n)})}{x_i^{(n)}} = \frac{\rho'(x_i^{(n)})}{x_i^{(n)}} \tag{11}$$

which leads to the Langevin simulation with step-size $\epsilon > 0$:

$$\mathbf{x}_{k+1} = \mathbf{x}_k + \frac{\epsilon}{2} \nabla \log p(\mathbf{x}_k) + \sqrt{\epsilon} \mathbf{z}_k = \mathbf{x}_k \left(1 - \frac{\epsilon}{2} \sum_{i=1}^N (\mathbf{F}^{(i)})^\top \mathbb{E}[\boldsymbol{\Omega}^{(i)}|\mathbf{x}_k] \mathbf{F}^{(i)} \mathbf{x}_k \right) + \sqrt{\epsilon} \mathbf{z}_k \tag{12}$$

where $\mathbf{z}_k \sim \mathcal{N}(0, \mathbf{I})$.

4 Diffusion for GGSM Models

Since the distribution of a GGSM model conditioned on the latent precision matrices $\boldsymbol{\Omega} = \{\boldsymbol{\Omega}^{(1)}, \dots, \boldsymbol{\Omega}^{(N)}\}$ is Gaussian with zero mean and precision matrix $\mathbf{P}(\boldsymbol{\Omega}) = \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)}$, the corresponding DDPM can be solved analytically without requiring a neural network to learn the score.

Under the standard DDPM forward noise schedule we have the transition $0 \rightarrow t$

$$q(\mathbf{x}_t | x_0) = \mathcal{N}(\mathbf{x}_t; \alpha_t \mathbf{x}_0, \sigma_t^2 \mathbf{I}) \tag{13}$$

where $\alpha_t = \sqrt{\bar{\alpha}_t}$ and $\sigma_t = \sqrt{1 - \bar{\alpha}_t}$ and the single-step forward transition $t-1 \rightarrow t$

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}) \tag{14}$$

where $\beta_t = 1 - \frac{\alpha_t^2}{\alpha_{t-1}^2}$ and for later use $\sigma_{t \rightarrow t-1} = \frac{\sigma_{t-1}}{\sigma_t} \sqrt{1 - \frac{\alpha_t^2}{\alpha_{t-1}^2}}$.

Conditioning on a specific realization of the latent scales $\boldsymbol{\Omega}$ gives the Gaussian noise-perturbed distribution

$$p(\mathbf{x}_t | \boldsymbol{\Omega}) = \mathcal{N}(\mathbf{0}, \alpha_t^2 \mathbf{P}(\boldsymbol{\Omega})^{-1} + \sigma_t^2 \mathbf{I}) \tag{15}$$

If the precision scales $\boldsymbol{\Omega}$ were fixed and known, estimating the clean data $\mathbf{x}_0$ given the noisy measurement $\mathbf{x}_t$ reduces to

$$\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t, \boldsymbol{\Omega}] = \arg \min_{\mathbf{x}_0} \left\{ \frac{1}{2\sigma_t^2} \|\mathbf{x}_t - \alpha_t \mathbf{x}_0\|^2 + \frac{1}{2} \sum_{n=1}^N \mathbf{x}_0^\top (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \mathbf{x}_0 \right\} \tag{16}$$

which produces the following closed-form conditional denoiser

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t, \boldsymbol{\Omega}] = \alpha_t \left(\alpha_t^2 \mathbf{I} + \sigma_t^2 \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \right)^{-1} \mathbf{x}_t \quad (17)$$

Now by the Law of Total Expectation, the marginal MMSE estimator for clean data $\mathbf{x}_0$ given noisy state $\mathbf{x}_t$ is

$$\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t] = \alpha_t \mathbb{E}_{\boldsymbol{\Omega} \mid \mathbf{x}_t} \left[\left(\alpha_t^2 \mathbf{I} + \sigma_t^2 \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \right)^{-1} \right] \mathbf{x}_t \quad (18)$$

Using Tweedie's formula, the exact score function $\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t)$ is

$$\nabla_{\mathbf{x}_t} \log p_t(\mathbf{x}_t) = -\frac{1}{\sigma_t^2} (\mathbf{x}_t - \alpha_t \mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t]) \quad (19)$$

To evaluate $\mathbb{E}[\mathbf{x}_0 \mid \mathbf{x}_t]$ in practice, the posterior expectation of the precision parameters $\hat{\boldsymbol{\Omega}}_t^{(n)} = \text{Diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$ is computed based on the choice of the prior distribution $\pi(\omega_i^{(n)})$ and an estimate of $\mathbf{x}_0$, $\hat{\mathbf{x}}_0$,

$$\hat{\omega}_i^{(n)} = \mathbb{E} \left[\omega_i^{(n)} \mid (\mathbf{F}^{(n)} \hat{\mathbf{x}}_0)_i \right] \quad (20)$$

Recall that from equation (11) we know the value of this expectation. That is, we can use a variational mean-field approximation.

The process works as follows, given as clean target estimate $\hat{\mathbf{x}}_0$, evaluate for each $n = 1, \dots, N$:

$$\hat{\boldsymbol{\Omega}}_t^{(n)} = \text{Diag} \left(\mathbb{E} \left[\boldsymbol{\Omega}^{(n)} \mid \mathbf{F}^{(n)} \hat{\mathbf{x}}_0 \right] \right) \quad (21)$$

then use the following approximation

$$\mathbb{E}_{\boldsymbol{\Omega} \mid \mathbf{x}_t} \left[\left(\alpha_t^2 \mathbf{I} + \sigma_t^2 \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \boldsymbol{\Omega}^{(n)} \mathbf{F}^{(n)} \right)^{-1} \right] \approx \left(\alpha_t^2 \mathbf{I} + \sigma_t^2 \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\boldsymbol{\Omega}}_t^{(n)} \mathbf{F}^{(n)} \right)^{-1} \quad (22)$$

For stability, we solve first

$$\left(\alpha_t^2 \mathbf{I} + \sigma_t^2 \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\boldsymbol{\Omega}}_t^{(n)} \mathbf{F}^{(n)} \right) \mathbf{s}_t = \mathbf{x}_t \quad (23)$$

and finally compute the updated estimate $\hat{\mathbf{x}}_0$ by solving, see equation (18),

$$\hat{\mathbf{x}}_0 = \alpha_t \mathbf{s}_t \quad (24)$$

Algorithm 1 implements the whole process

Algorithm 1 Score-Based DDPM Sampling for GGSM Models

```
1: Input: Transform operators  $\{\mathbf{F}^{(n)}\}_{n=1}^N$ , noise schedule parameters  $\{\beta_t, \alpha_t, \sigma_t, \sigma_{t \rightarrow t-1}\}_{t=1}^T$ ,  
   prior posterior expectation functions  $\mathbb{E}[\omega \mid \cdot]$ , total steps  $T$ , variational mean field iterations  
    $M$ .  
2:  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ ,  $\hat{\mathbf{x}}_0 \leftarrow \mathbf{0}$ . ▷ noise sample and target estimate initialization  
3: for  $t = T, T-1, \dots, 1$  do  
4:   for  $m = 1, \dots, M$  do ▷ Iterative precision estimation loop  
5:     for  $n = 1, \dots, N$  do  
6:        $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \hat{\mathbf{x}}_0$  ▷ Compute transform output  
7:        $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} \mid r_i^{(n)}] \quad \forall i = 1, \dots, D$  ▷ Update diagonal precision values  
8:        $\hat{\Omega}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$  ▷ Form precision matrix component  
9:     end for  
10:     $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\Omega}^{(n)} \mathbf{F}^{(n)}$  ▷ Reconstruct total precision  
11:     $\mathbf{v}_t \leftarrow (\alpha_t^2 \mathbf{I} + \sigma_t^2 \mathbf{P})^{-1} \mathbf{x}_t$  ▷ Solve linear system for auxiliary vector  
12:     $\hat{\mathbf{x}}_0 \leftarrow \alpha_t \mathbf{v}_t$  ▷ Update clean target estimate  
13:  end for  
14:   $\mathbf{s}_p \leftarrow -\mathbf{P} \mathbf{v}_t$  [equivalently  $\mathbf{s}_p \leftarrow -\frac{1}{\sigma_t^2} (\mathbf{x}_t - \alpha_t \hat{\mathbf{x}}_0)$ ] ▷ Compute Score Vector  
15:  if  $t > 1$  then  
16:     $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  ▷ Draw Gaussian noise  
17:  else  
18:     $\mathbf{z} \leftarrow \mathbf{0}$  ▷ Set noise  
19:  end if  
20:   $\mathbf{x}_{t-1} \leftarrow \frac{1}{\sqrt{1-\beta_t}} (\mathbf{x}_t + \beta_t \mathbf{s}_p) + \sigma_{t \rightarrow t-1} \mathbf{z}$  ▷ Execute Reverse Score Step:  
21: end for  
22: Generated sample  $\mathbf{x}_0$  ▷ Output
```

5 Integration of GGSM and Generative Diffusion Models

The next step in the evolution of our GGSM model is to combine it with a probability distribution for which, using only samples, we have learned a diffusion model.

We so consider simulating from a normalized tempered density defined by

$$r(\mathbf{x}) = C_\lambda^{-1} q^\lambda(\mathbf{x}) p^{1-\lambda}(\mathbf{x}) \quad (25)$$

where $\lambda \in (0, 1)$, and C_λ is the unknown normalizing constant.

Maybe we should replace λ and $1 - \lambda$ by positive independent numbers

Taking the gradient of the logarithm of $r(\mathbf{x})$ produces

$$\nabla_{\mathbf{x}} \log r(\mathbf{x}) = \lambda \nabla_{\mathbf{x}} \log q(\mathbf{x}) - (1 - \lambda) \sum_{i=1}^N (\mathbf{F}^{(i)})^\top \mathbb{E}[\Omega^{(i)} | \mathbf{x}^{(i)}] \mathbf{F}^{(i)} \mathbf{x} \quad (26)$$

When the density $q(\mathbf{x})$ belongs to a data distribution represented by a trained diffusion model, the score network $\mathbf{s}_\theta(\mathbf{x}, t) \approx \nabla \log q_t(\mathbf{x})$ can be integrated statically or dynamically with our GGSM distribution $p(\mathbf{x})$.

Two approaches are possible. Frist, to evaluate the score network at a minimal noise boundary

$t = \epsilon_t \ll 1$. This provides a static approximation

$$\nabla_{\mathbf{x}} \log r(\mathbf{x}) \approx \lambda \mathbf{s}_\theta(\mathbf{x}, \epsilon_t) - (1 - \lambda) \sum_{i=1}^N (\mathbf{F}^{(n)})^\top \mathbb{E}[\boldsymbol{\Omega}^{(n)} | \mathbf{x}^{(n)}] \mathbf{F}^{(n)} \mathbf{x} \quad (27)$$

That is, we only introduce the GGSM model at the late state of the diffusion.

Alternatively, we can also perform dynamic sampling across the geometric mixture trajectory $\mathbf{r}_t(x_t) \propto q_t^\lambda(x_t) p_t^{1-\lambda}(x_t)$, the composite score gradient is unified at each timestep:

$$\nabla_{x_t} \log \mathbf{r}_t(x_t) = \lambda \mathbf{s}_\theta(x_t, t) - \frac{1 - \lambda}{\sigma_t^2} (\mathbf{x}_t - \alpha_t \mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]) \quad (28)$$

where $\mathbb{E}[\mathbf{x}_0 | \mathbf{x}_t]$ is calculated for the GGSM model, see section 4.

The following algorithm, Algorithm 2, samples from the tempered distribution.

Algorithm 2 Stochastic Reverse SDE Geometric Sampling

```

1: Input: Transform operators  $\{\mathbf{F}^{(n)}\}_{n=1}^N$ , noise schedule parameters  $\{\beta_t, \alpha_t, \sigma_t, \sigma_{t \rightarrow t-1}\}_{t=1}^T$ ,
   prior posterior expectation functions  $\mathbb{E}[\omega | \cdot]$ , total steps  $T$ , variational mean field iterations,
    $M$ , score network  $\mathbf{s}_\theta(\mathbf{x}_t, t)$ , blend  $\lambda$ .
2: Sample noise  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , ▷ Initialization
3: for  $t = T, T-1, \dots, 1$  do
4:    $\mathbf{s}_q \leftarrow \mathbf{s}_\theta(\mathbf{x}_t, t)$  ▷ score from trained diffusion model
5:    $\hat{\mathbf{x}}_0 \leftarrow \frac{1}{\alpha_t} (\mathbf{x}_t + \sigma_t^2 \mathbf{s}_q)$  ▷ clean image initialization
6:   for  $m = 1, \dots, M$  do ▷ Iterative precision estimation loop
7:     for  $n = 1, \dots, N$  do
8:        $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \hat{\mathbf{x}}_0$  ▷ Compute transform output
9:        $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} | r_i^{(n)}] \quad \forall i = 1, \dots, D$  ▷ Update diagonal precision values
10:       $\hat{\boldsymbol{\Omega}}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$  ▷ Form precision matrix component
11:    end for
12:     $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\boldsymbol{\Omega}}^{(n)} \mathbf{F}^{(n)}$  ▷ Reconstruct total precision
13:     $\mathbf{v}_p \leftarrow (\alpha_t^2 \mathbf{I} + \sigma_t^2 \mathbf{P})^{-1} \mathbf{x}_t$  ▷ Solve linear system for auxiliary vector
14:     $\hat{\mathbf{x}}_0 \leftarrow \alpha_t \mathbf{v}_p$  ▷ Update clean target estimate
15:  end for
16:   $\mathbf{s}_p \leftarrow -\mathbf{P} \mathbf{v}_p$  ▷ Compute Score Vector
   [equivalently  $\mathbf{s}_p \leftarrow -\frac{1}{\sigma_t^2} (\mathbf{x}_t - \alpha_t \hat{\mathbf{x}}_0)$ ]
17:   $\mathbf{s}_r \leftarrow \lambda \mathbf{s}_q + (1 - \lambda) \mathbf{s}_p$ 
18:  if  $t > 1$  then
19:     $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  ▷ Draw Gaussian noise
20:  else
21:     $\mathbf{z} \leftarrow \mathbf{0}$  ▷ Set noise
22:  end if
23:   $\mathbf{x}_{t-1} \leftarrow \frac{1}{\sqrt{1-\beta_t}} (\mathbf{x}_t + \beta_t \mathbf{s}_r) + \sigma_{t \rightarrow t-1} \mathbf{z}$  ▷ Execute Reverse Score Step:
24: end for
25: Generated sample  $\mathbf{x}_0$  ▷ Output

```

The following algorithm, Algorithm 3, mirrors the marginal probability flow mapping exactly onto a DDIM-style trajectory, omitting stochastic updates.

Algorithm 3 Deterministic Probability Flow ODE Geometric Sampling

```
1: Input: Transform operators  $\{\mathbf{F}^{(n)}\}_{n=1}^N$ , noise schedule parameters  $\{\beta_t, \alpha_t, \sigma_t, \sigma_{t \rightarrow t-1}\}_{t=1}^T$ ,  
prior posterior expectation functions  $\mathbb{E}[\omega \mid \cdot]$ , total steps  $T$ , variational mean field iterations,  
 $M$ , score network  $\mathbf{s}_\theta(\mathbf{x}_t, t)$ , blend  $\lambda$ .  
2: Sample noise  $\mathbf{x}_T \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , ▷ Initialization  
3: for  $t = T, T-1, \dots, 1$  do  
4:    $\mathbf{s}_q \leftarrow \mathbf{s}_\theta(\mathbf{x}_t, t)$  ▷ score from trained diffusion model  
5:    $\hat{\mathbf{x}}_0 \leftarrow \frac{1}{\alpha_t}(\mathbf{x}_t + \sigma_t^2 \mathbf{s}_q)$  ▷ clean image initialization  
6:   for  $m = 1, \dots, M$  do ▷ Iterative precision estimation loop  
7:     for  $n = 1, \dots, N$  do  
8:        $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \hat{\mathbf{x}}_0$  ▷ Compute transform output  
9:        $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} \mid r_i^{(n)}] \quad \forall i = 1, \dots, D$  ▷ Update diagonal precision values  
10:       $\hat{\Omega}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$  ▷ Form precision matrix component  
11:    end for  
12:     $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\Omega}^{(n)} \mathbf{F}^{(n)}$  ▷ Reconstruct total precision  
13:     $\mathbf{v}_t \leftarrow (\alpha_t^2 \mathbf{I} + \sigma_t^2 \mathbf{P})^{-1} \mathbf{x}_t$  ▷ Solve linear system for auxiliary vector  
14:     $\hat{\mathbf{x}}_0 \leftarrow \alpha_t \mathbf{v}_t$  ▷ Update clean target estimate  
15:  end for  
16:   $\mathbf{s}_p \leftarrow -\mathbf{P} \mathbf{v}_t$  [equivalently  $\mathbf{s}_p \leftarrow -\frac{1}{\sigma_t^2}(\mathbf{x}_t - \alpha_t \hat{\mathbf{x}}_0)$ ] ▷ Compute Score Vector  
17:   $\mathbf{s}_{\text{total}} \leftarrow \lambda \mathbf{s}_q + (1 - \lambda) \mathbf{s}_p$   
18:   $\hat{\mathbf{x}}_{0, \text{total}} \leftarrow \frac{1}{\alpha_t}(\mathbf{x}_t + \sigma_t^2 \mathbf{s}_{\text{total}})$   
19:   $\mathbf{x}_{t-1} \leftarrow \alpha_{t-1} \hat{\mathbf{x}}_{0, \text{total}} - \sigma_{t-1} \sigma_t \mathbf{s}_{\text{total}}$   
20: end for  
21: Generated sample  $\mathbf{x}_0$  ▷ Output
```

6 Flow Matching for Generalized Gaussian Scale Mixture Models

We now study the Flow Matching (FM) simulation of GGSM models and tempered geometric distributions as we did for diffusion in the previous two sections. Although this study can be derived from this previous two sections (we are using Gaussian paths) we prefer to develop it independently for clarify.

For the Generalized Gaussian Scale Mixture (GGSM) model defined by N linear transform operators $\{\mathbf{F}^{(n)}\}_{n=1}^N$:

$$p(\mathbf{x}_1 \mid \Omega) \propto \mathcal{N}(\mathbf{x}_1 \mid \mathbf{0}, \mathbf{P}^{-1}(\Omega)), \quad \mathbf{P}(\Omega) = \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \Omega^{(n)} \mathbf{F}^{(n)} \quad (29)$$

the Optimal Transport Flow Matching trajectory integrates forward in time from $t = 0$ (pure standard Gaussian noise $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$) to $t = 1$ (clean generated sample $\mathbf{x}_1$) via the Ordinary Differential Equation (ODE)

$$\frac{d\mathbf{x}_t}{dt} = \mathbf{v}_t(\mathbf{x}_t) \quad (30)$$

Let's assume we have an estimate, $\hat{\Omega}$, of Ω . In Optimal Transport Flow Matching, the trajectory is the linear interpolation:

$$\mathbf{x}_t = (1 - t)\mathbf{x}_0 + t\mathbf{x}_1 \quad (31)$$

which produces

$$\mathbf{x}_1 - \mathbf{x}_0 = \frac{\mathbf{x}_1 - \mathbf{x}_0}{1 - t} \quad (32)$$

Taking the conditional expectation gives:

$$\mathbf{v}_t(\mathbf{x}_t | \hat{\mathbf{\Omega}}) = \mathbb{E}[\mathbf{x}_1 - \mathbf{x}_0 | \mathbf{x}_t, \hat{\mathbf{\Omega}}] = \frac{\mathbb{E}[\mathbf{x}_1 | \mathbf{x}_t, \hat{\mathbf{\Omega}}] - \mathbf{x}_t}{1 - t} \quad (33)$$

Now, since $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ and $\mathbf{x}_1 | \hat{\mathbf{\Omega}} \sim \mathcal{N}(\mathbf{0}, \mathbf{P}^{-1}(\hat{\mathbf{\Omega}}))$ are independent zero-mean Gaussians we have

$$\text{Cov}(\mathbf{x}_1, \mathbf{x}_t | \hat{\mathbf{\Omega}}) = \text{Cov}(\mathbf{x}_1, (1 - t)\mathbf{x}_0 + t\mathbf{x}_1 | \hat{\mathbf{\Omega}}) = t\mathbf{P}^{-1}(\hat{\mathbf{\Omega}}) \quad (34)$$

$$\text{Cov}(\mathbf{x}_t, \mathbf{x}_t | \hat{\mathbf{\Omega}}) = (1 - t)^2\mathbf{I} + t^2\mathbf{P}^{-1}(\hat{\mathbf{\Omega}}) \quad (35)$$

From which we obtain

$$\begin{aligned} \mathbb{E}[\mathbf{x}_1 | \mathbf{x}_t, \hat{\mathbf{\Omega}}] &= \text{Cov}(\mathbf{x}_1, \mathbf{x}_t | \hat{\mathbf{\Omega}}) \left[\text{Cov}(\mathbf{x}_t, \mathbf{x}_t | \hat{\mathbf{\Omega}}) \right]^{-1} \mathbf{x}_t \\ &= t\mathbf{P}^{-1}(\hat{\mathbf{\Omega}}) \left((1 - t)^2\mathbf{I} + t^2\mathbf{P}^{-1}(\hat{\mathbf{\Omega}}) \right)^{-1} \mathbf{x}_t \\ &= t \left(t^2\mathbf{I} + (1 - t)^2\mathbf{P}(\hat{\mathbf{\Omega}}) \right)^{-1} \mathbf{x}_t \\ &= t\mathbf{w}_t \end{aligned} \quad (36)$$

where

$$\mathbf{w}_t = \left(t^2\mathbf{I} + (1 - t)^2\mathbf{P}(\hat{\mathbf{\Omega}}) \right)^{-1} \mathbf{x}_t \quad (37)$$

which produces

$$\mathbf{v}_t(\mathbf{x}_t | \hat{\mathbf{\Omega}}) = \frac{t\mathbf{w}_t - \mathbf{x}_t}{1 - t} = \left(t\mathbf{I} - (1 - t)\mathbf{P}(\hat{\mathbf{\Omega}}) \right) \mathbf{w}_t \quad (38)$$

The auxiliary vector $\mathbf{w}_t$ simultaneously exposes the MMSE prediction of the clean target state $\hat{\mathbf{x}}_1$, which is used inside a fixed-point loop to dynamically update the precision estimates $\hat{\mathbf{\Omega}}$:

$$\hat{\mathbf{x}}_1 = t\mathbf{w}_t \quad (39)$$

Algorithm 4 implements the proposed Flow Matching model

Algorithm 4 Optimal Transport Flow Matching ODE Sampler for GGSM

- 1: **Input:** Transform operators $\{\mathbf{F}^{(n)}\}_{n=1}^N$, prior posterior expectation functions $\mathbb{E}[\omega | \cdot]$, total integration ODE steps K , inner fixed-point iterations M .
 - 2: **Initialize:** Set step size $\Delta t \leftarrow 1/K$, sample noise $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, initialize baseline target estimate $\hat{\mathbf{x}}_1 \leftarrow \mathbf{0}$.
 - 3: **for** $k = 0, 1, \dots, K - 1$ **do**
 - 4: $t \leftarrow k \cdot \Delta t$ ▷ Set current continuous time
 - 5: **for** $m = 1, \dots, M$ **do** ▷ Iterative precision estimation loop
 - 6: **for** $n = 1, \dots, N$ **do**
 - 7: $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)}\hat{\mathbf{x}}_1$ ▷ Compute transform output
 - 8: $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} | r_i^{(n)}] \quad \forall i = 1, \dots, D$ ▷ Update diagonal precision values
 - 9: $\hat{\mathbf{\Omega}}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$ ▷ Form precision component
 - 10: **end for**
 - 11: $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\mathbf{\Omega}}^{(n)} \mathbf{F}^{(n)}$ ▷ Reconstruct total precision
 - 12: $\mathbf{w}_t \leftarrow (t^2\mathbf{I} + (1 - t)^2\mathbf{P})^{-1} \mathbf{x}_t$ ▷ Solve linear system for auxiliary vector
 - 13: $\hat{\mathbf{x}}_1 \leftarrow t\mathbf{w}_t$ ▷ Update clean target estimate
 - 14: **end for**
 - 15: $\mathbf{v}_t \leftarrow t\mathbf{w}_t - (1 - t)\mathbf{P}\mathbf{w}_t$ ▷ Compute Velocity Vector
 - 16: $\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t + \Delta t \cdot \mathbf{v}_t$ ▷ Execute Forward ODE Step (Euler)
 - 17: **end for**
 - 18: **Output:** Generated sample $\mathbf{x}_1$
-

7 Integration of GGSM and Flow Matching Models

One of the properties of the FM framework is how it handles geometric density mixtures. Because Flow Matching focuses on predicting a velocity field, translating a geometric density mixture into a flow matching formulation yields a clean mathematical shortcut: The total velocity field of a geometric mixture is the linear interpolation of the individual components' velocity fields, as in diffusion is of individual scores.

Whether we choose a curved trajectory or a straight trajectory, the composite velocity field $v_{\mathbf{r}}(x_t, t)$ for the geometric mixture $\mathbf{r}(x) \propto q^\lambda(x)p^{1-\lambda}(x)$ resolves directly to

$$\mathbf{v}_{\mathbf{r}}(x_t, t) = \lambda \mathbf{v}_{\theta}(x_t, t) + (1 - \lambda) \mathbf{v}_p(x_t, t) \quad (40)$$

where $\mathbf{v}_{\theta}(x_t, t)$ is the trained data neural network, and $\mathbf{v}_p(x_t, t)$ is the analytical velocity field of the GGSM prior.

Using the results from section (6), Algorithm (5) shows the process to obtain samples from a geometric mixture model.

Algorithm 5 OT-FM ODE Sampler for GGSM Geometric Mixtures

```

1: Input: Base velocity model  $\mathbf{v}_{\theta}(\cdot, t)$ , transform operators  $\{\mathbf{F}^{(n)}\}_{n=1}^N$ , prior posterior functions  $\mathbb{E}[\omega | \cdot]$ , mixture weight  $\lambda \in [0, 1]$ , total ODE steps  $K$ , fixed-point inner iterations  $M$ .
2: Initialize: Step size  $\Delta t \leftarrow 1/K$ , noise sample  $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ , baseline target  $\hat{\mathbf{x}}_{1,r} \leftarrow \mathbf{0}$ .
3: for  $k = 0, 1, \dots, K - 1$  do
4:    $t \leftarrow k \cdot \Delta t$  ▷ Set continuous time
5:    $\mathbf{v}_{\mathbf{q}} \leftarrow \mathbf{v}_{\theta}(\mathbf{x}_t, t)$  ▷ velocity from trained FM model
6:   for  $m = 1, \dots, M$  do ▷ Iterative GGSM precision refinement
7:     for  $n = 1, \dots, N$  do
8:        $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \hat{\mathbf{x}}_{1,r}$  ▷ Compute transform output
9:        $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} | r_i^{(n)}] \quad \forall i = 1, \dots, D$  ▷ Update diagonal precision values
10:       $\hat{\Omega}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$  ▷ Form precision component
11:    end for
12:     $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\Omega}^{(n)} \mathbf{F}^{(n)}$  ▷ Reconstruct GGSM total precision
13:     $\mathbf{w}_p \leftarrow (t^2 \mathbf{I} + (1 - t)^2 \mathbf{P})^{-1} \mathbf{x}_t$  ▷ Solve linear system for auxiliary vector
14:     $\mathbf{v}_p \leftarrow t \mathbf{w}_p - (1 - t) \mathbf{P} \mathbf{w}_p$  ▷ Compute GGSM velocity component:
15:     $\mathbf{v}_{\mathbf{r}} \leftarrow \lambda \mathbf{v}_{\mathbf{q}} + (1 - \lambda) \mathbf{v}_p$  ▷ Compute mixture velocity field
16:     $\hat{\mathbf{x}}_{1,r} \leftarrow \mathbf{x}_t + (1 - t) \mathbf{v}_{\mathbf{r}}$  ▷ Update predicted clean target
17:  end for
18:   $\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t + \Delta t \cdot \mathbf{v}_{\mathbf{r}}$  ▷ Execute Forward ODE Integration Step (Euler)
19: end for
20: Generated sample  $\mathbf{x}_1 \sim r(\mathbf{x})$  ▷ Output

```

8 Plug and Play Restoration for Geometric Density Mixture Flow Matching

We now describe a hybrid generative path governed by three distinct forces under the Optimal Transport Flow Matching (OT-FM) framework. We have a learned Data Prior ($\mathbf{v}_{\theta}$) which captures complex high-dimensional semantic data distributions, an analytical GGSM Prior ($\mathbf{v}_p$) which imposes structural regularizations (e.g., edge or wavelet sparsity constraints), and finally a data fidelity penalty ($\mathbf{F}(\mathbf{x})$) which enforces physical consistency with the observed degraded target $\mathbf{y}$. This terms has the form $\mathbf{F}(\mathbf{x}) = \frac{1}{2\sigma^2} \|\mathbf{y} - \mathcal{A}(\mathbf{x})\|^2$ where, for the time being ($\mathcal{A}$) is a blurring matrix.

The unconditional prior field is formulated as a linear mixture:

$$\mathbf{v}_r(\mathbf{x}) = \lambda \mathbf{v}_\theta(\mathbf{x}) + (1 - \lambda) \mathbf{v}_p(\mathbf{x}) \quad (41)$$

To avoid expensive backpropagation through the neural network parameters during restoration, a Plug-and-Play (PnP) operator splitting mechanism updates the composite clean target prediction ($\hat{\mathbf{x}}_1$) via direct gradient descent on $\mathbf{F}(\mathbf{x})$ at each step along the continuous time trajectory $t \in [0, 1]$. However, I'm not 100% sure that the order of all the sentences is the correct one. We should try alternative orders.

Algorithm (6) contains a first possible restoration method.

Algorithm 6 Unified OT-FM PnP Geometric Restoration

- 1: **Input:** Base velocity model $\mathbf{v}_\theta(\cdot, t)$, transform operators $\{\mathbf{F}^{(n)}\}_{n=1}^N$, prior posterior functions $\mathbb{E}[\omega \mid \cdot]$, mixture weight $\lambda \in [0, 1]$, total ODE steps K , fixed-point inner iterations M , Time-dependent gradient step-size schedule for the fidelity penalty γ_t , observed image $\mathbf{y}$.
 - 2: **Initialize:** Step size $\Delta t \leftarrow 1/K$, noise sample $\mathbf{x}_0 \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$, baseline target $\hat{\mathbf{x}}_{1,r}$, initial restoration.
 - 3: **for** $k = 0, 1, \dots, K - 1$ **do**
 - 4: $t \leftarrow k \cdot \Delta t$ ▷ Set continuous time
 - 5: $\mathbf{v}_q \leftarrow \mathbf{v}_\theta(\mathbf{x}_t, t)$ ▷ velocity from trained FM model
 - 6: **for** $m = 1, \dots, M$ **do** ▷ Iterative GGSM precision refinement
 - 7: **for** $n = 1, \dots, N$ **do**
 - 8: $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \hat{\mathbf{x}}_{1,r}$ ▷ Compute transform output
 - 9: $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E}[\omega_i^{(n)} \mid r_i^{(n)}] \quad \forall i = 1, \dots, D$ ▷ Update diagonal precision values
 - 10: $\hat{\Omega}^{(n)} \leftarrow \text{diag}(\hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)})$ ▷ Form precision component
 - 11: **end for**
 - 12: $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\Omega}^{(n)} \mathbf{F}^{(n)}$ ▷ Reconstruct GGSM total precision
 - 13: $\mathbf{w}_p \leftarrow (t^2 \mathbf{I} + (1 - t)^2 \mathbf{P})^{-1} \mathbf{x}_t$ ▷ Solve linear system for auxiliary vector
 - 14: $\mathbf{v}_p \leftarrow t \mathbf{w}_p - (1 - t) \mathbf{P} \mathbf{w}_p$ ▷ Compute GGSM velocity component:
 - 15: $\mathbf{v}_r \leftarrow \lambda \mathbf{v}_q + (1 - \lambda) \mathbf{v}_p$ ▷ Compute mixture velocity field
 - 16: $\hat{\mathbf{x}}_{1,r} \leftarrow \mathbf{x}_t + (1 - t) \mathbf{v}_r$ ▷ Update predicted clean target
 - 17: **end for**
 - 18: $\hat{\mathbf{x}}_1^* \leftarrow \hat{\mathbf{x}}_{1,r} - \gamma_t \nabla \mathbf{F}(\hat{\mathbf{x}}_{1,r})$ ▷ Steer target via PnP fidelity step
 - 19: $\hat{\mathbf{x}}_0 \leftarrow \mathbf{x}_t - t \mathbf{v}_r$ ▷ Isolate current latent path noise
 - 20: $\mathbf{x}_t^* \leftarrow t \hat{\mathbf{x}}_1^* + (1 - t) \hat{\mathbf{x}}_0$ ▷ Reproject onto the corrected flow coordinate
 - 21: $\mathbf{x}_{t+\Delta t} \leftarrow \mathbf{x}_t^* + \Delta t \cdot \mathbf{v}_r$ ▷ Advance forward to next time horizon
 - 22: **end for**
 - 23: **Output** $\mathbf{x}_1$ ▷ Fully restored target image
-

Finally, we can also utilize the approach in the paper entitled *PnP-Flow: PLUG-AND-PLAY Image Restoration with Flow Matching*. Algorithm (7) describes how to do it.

Algorithm 7 Unified OT-FM PnP Geometric Restoration

```

1: Input: Base velocity model  $\mathbf{v}_\theta(\cdot, t)$ , transform operators  $\{\mathbf{F}^{(n)}\}_{n=1}^N$ , prior posterior functions
    $\mathbb{E}[\omega \mid \cdot]$ , mixture weight  $\lambda \in [0, 1]$ , fixed-point inner iterations  $M$ ,  $t_k = k/K, K \in \mathbb{N}$ , step
   size  $\gamma_k$ , observed image  $\mathbf{y}$ .
2: Initialize:  $\mathbf{x}_0$ , initial restoration.
3: for  $k = 0, 1, \dots, K$  do
4:    $\mathbf{z}_k \leftarrow \mathbf{x}_k - \gamma_k \nabla \mathbf{F}(\mathbf{x}_k)$  ▷ Steer target via PnP fidelity step
5:    $\tilde{\mathbf{z}}_k \leftarrow (1 - t_k)\epsilon + t_k \mathbf{z}_k, \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$  ▷ Linear interpolation
6:    $\mathbf{v}_q \leftarrow \mathbf{v}_\theta(\tilde{\mathbf{z}}_k, t_k)$  ▷ velocity from trained FM model
7:   for  $m = 1, \dots, M$  do ▷ Iterative GGSM precision refinement
8:     for  $n = 1, \dots, N$  do
9:        $\mathbf{r}^{(n)} \leftarrow \mathbf{F}^{(n)} \mathbf{x}_k$  ▷ Compute transform output
10:       $\hat{\omega}_i^{(n)} \leftarrow \mathbb{E} \left[ \omega_i^{(n)} \mid r_i^{(n)} \right] \quad \forall i = 1, \dots, D$  ▷ Update diagonal precision values
11:       $\hat{\Omega}^{(n)} \leftarrow \text{diag} \left( \hat{\omega}_1^{(n)}, \dots, \hat{\omega}_D^{(n)} \right)$  ▷ Form precision component
12:    end for
13:     $\mathbf{P} \leftarrow \sum_{n=1}^N (\mathbf{F}^{(n)})^\top \hat{\Omega}^{(n)} \mathbf{F}^{(n)}$  ▷ Reconstruct GGSM total precision
14:     $\mathbf{w}_p \leftarrow (t^2 \mathbf{I} + (1 - t)^2 \mathbf{P})^{-1} \mathbf{x}_k$  ▷ Solve linear system for auxiliary vector
15:     $\mathbf{v}_p \leftarrow t \mathbf{w}_p - (1 - t) \mathbf{P} \mathbf{w}_p$  ▷ Compute GGSM velocity component:
16:     $\mathbf{v}_r \leftarrow \lambda \mathbf{v}_q + (1 - \lambda) \mathbf{v}_p$  ▷ Compute mixture velocity field
17:  end for
18:   $\mathbf{x}_{k+1} \leftarrow \mathbf{D}_{t_k}(\tilde{\mathbf{z}}_k) = \tilde{\mathbf{z}}_k + (1 - t_k) \mathbf{v}_r$  ▷ PnD step with denoiser
19: end for
20: Output  $\mathbf{x}_{n+1}$  ▷ Fully restored target image

```
